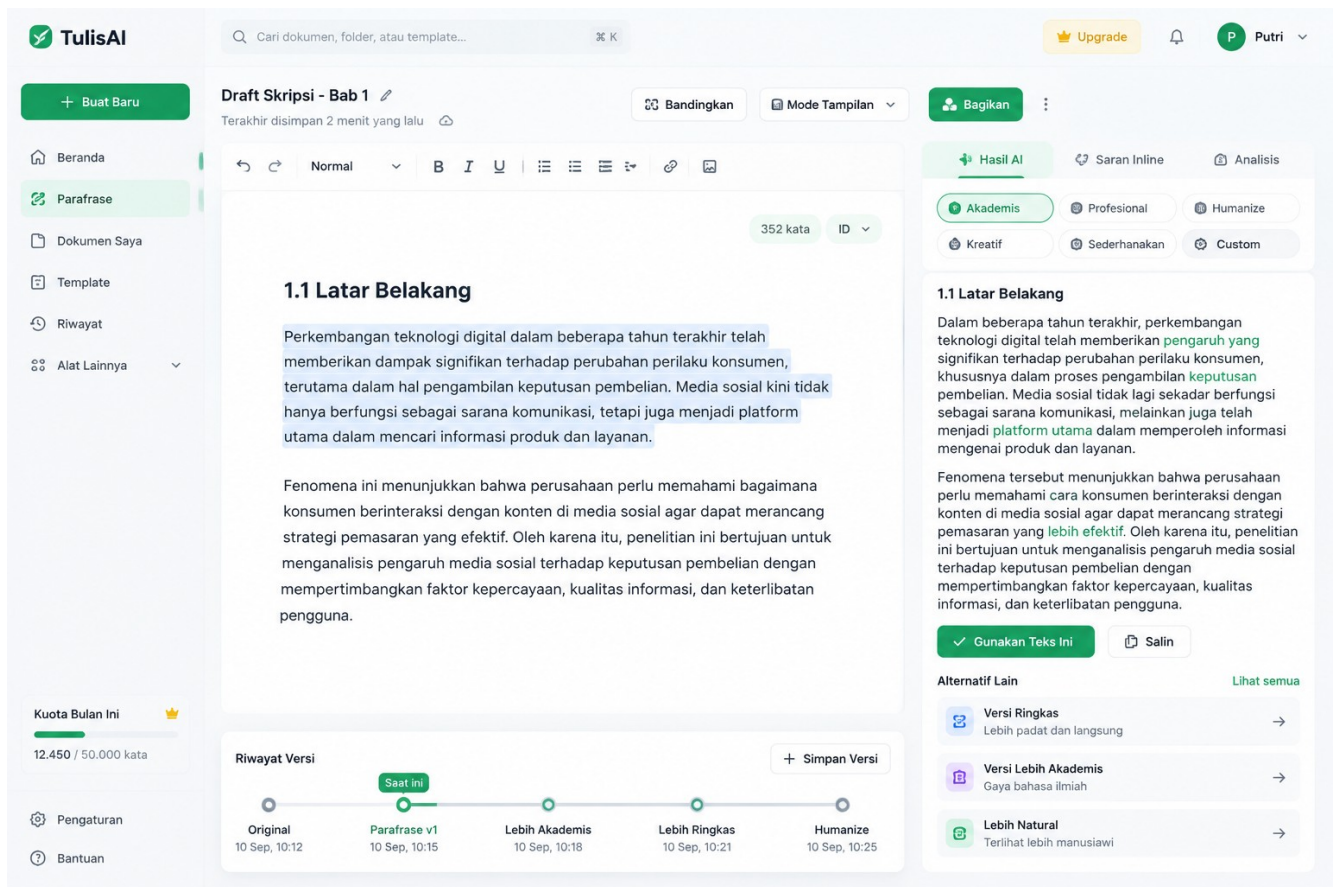


PRODUCT REQUIREMENTS DOCUMENT

AI Writing Workspace

Bilingual Paraphraser & Writing Assistant

Prioritas Bahasa Indonesia



Visual direction only - final UI follows the interaction specification in this PRD.

PRD Version 1.0 | Product Name: TBD | Primary AI Model: GPT-5.6 Luna

0. Document Status, Scope, and Locked Decisions

This PRD is the consolidated product blueprint from the full product discussion. It intentionally separates locked product decisions from items that still require later research, especially AI system prompts. The document is written to be usable by product, UI/UX, frontend, backend, and AI engineering teams without requiring them to reconstruct the conversation history.

Decision	Status	Final direction
Product category	LOCKED	AI Writing Workspace, not a one-shot paraphraser wrapper.
Primary market	LOCKED	Indonesia first; students/thesis users and academics are primary, professionals and creators are secondary.
Languages	LOCKED	Bilingual Bahasa Indonesia + English; Indonesian UX and writing quality are the priority.
Editor paradigm	LOCKED	Google Docs-like document workspace after generation; single main canvas, not permanent side-by-side input/output boxes.
Entry experience	LOCKED	Simple centered quick-start input inspired by familiar paraphraser tools; advanced controls are progressively disclosed.
Writing modes	LOCKED	Standard/Paraphrase, Academic, Humanize, Professional, Creative, Simplify.
Custom instruction UX	LOCKED	Separate structured customization feature; user should not need to write a long prompt.
Inline editing	LOCKED	Selection-level rewrite and phrase alternatives; avoid full regenerate when only a small span needs change.
Term protection	LOCKED	User can lock technical terms; Academic mode also protects recognized citations.
Version control	LOCKED	Git-like safety expressed as user-friendly versions/timeline. Original is preserved.
Diff	LOCKED	On-demand compare/diff, not permanent split view.
Analytics	LOCKED	Fast local metrics + optional AI qualitative writing analysis.
AI architecture	LOCKED	Hybrid: AI only for semantic language tasks; deterministic logic for diff, history, counts, formatting, validation.
Primary model	LOCKED	GPT-5.6 Luna for MVP.
System prompts	TBD BY USER	Prompt content is intentionally left blank in this PRD; only prompt IDs, inputs, outputs, and validation are defined.
Plagiarism / AI detector	OUT OF MVP	Not required for initial differentiation and can introduce cost/false confidence.

Core product promise

The product must give users better text while keeping them in control: clear intent selection, predictable customization, protected meaning/terms, transparent differences, and reversible versions.

1. Executive Summary

The product starts from a crowded category - AI paraphrasing - but deliberately avoids competing only on “paste text -> generate rewrite.” The differentiator is the editing workflow around the model. Users should be able to tell the system what kind of writing they need, shape the output without learning prompt engineering, inspect exactly what changed, protect terms and citations, refine only selected passages, and safely move between versions.

The primary experience is therefore a document-centric writing workspace. The first interaction remains extremely simple: paste or type text, choose a writing intent, optionally customize the result, and generate. After that first transformation, the experience opens into a Google Docs-like editor with an AI assistant panel, inline actions, a version timeline, diff/compare mode, and writing analytics.

The architecture is intentionally cost-aware. Word count, reading time, diff, change percentage, formatting, autosave, version history, citation pattern detection, and locked-term validation do not need an LLM. GPT-5.6 Luna is reserved for tasks that genuinely require semantic understanding: paraphrasing, humanization, audience adaptation, meaningful shortening/expansion, semantic bulletization, selected-text alternatives, and optional qualitative evaluation.

North-star experience

A non-technical Indonesian user should be able to get a high-quality, context-appropriate rewrite without writing a prompt, while a power user should be able to refine individual words, phrases, paragraphs, and versions without repeatedly regenerating the entire document.

1.1 Product positioning

Do not position as	Position as
“AI spinner” / generic paraphraser	AI Writing Workspace for Indonesian writing refinement
“Beat AI detectors”	Make writing more natural and context-appropriate
One-shot generator	Editable, reversible, inspectable writing workflow
English tool translated into Indonesian	Indonesia-first writing behavior and UX, with English support

1.2 Product success definition

- Users reach a useful first result with minimal configuration.
- Applied AI changes preserve locked terms and recognized citations.
- Users can understand what changed through diff and change summaries.
- Users can undo/restore any AI transformation without losing the original.
- Humanizer outputs sound more natural without introducing grammar degradation, fake facts, or deliberately inserted errors.
- LLM calls occur only when semantic language transformation or qualitative evaluation is required.

2. Users, Jobs-to-be-Done, and Pain Points

Segment	Primary jobs	Pain points the product must solve
Student / thesis writer	Improve academic wording, shorten or clarify paragraphs, preserve variables/theory/citations, make AI-assisted drafts sound natural.	Does not know prompt engineering; afraid meaning changes; repetitive regenerations; technical terms get altered; hard to compare versions.
Academic / researcher	Refine clarity/formality, produce journal-appropriate wording, preserve citations and domain terms.	Needs precision, traceability, minimal semantic drift, and control over language register.
Professional	Polish email, report, proposal, presentation copy.	Needs concise, audience-appropriate output with fast turnaround.
Content / marketing user	Creative rewrite, story/marketing variants, natural copy.	Wants more expressive variation without rewriting everything manually.

2.1 Core jobs-to-be-done

1. “When my paragraph is understandable but not good enough, help me improve it without changing the intended meaning.”
2. “When I know the outcome I want but not how to write an AI prompt, give me structured controls that express my request.”
3. “When AI changed too much, let me fix only the problematic phrase instead of regenerating the whole paragraph.”
4. “When I use academic terms or citations, guarantee that they are not silently rewritten.”
5. “When I try several rewrites, keep all meaningful versions so I can compare and restore safely.”
6. “When I humanize AI-assisted text, make it genuinely natural for the context - not intentionally worse or full of mistakes.”

2.2 Primary user anxieties to design around

- “Did the AI change my meaning?” -> diff + conservative controls + version restore.
- “Did it change my theory/variable/citation?” -> term locks + citation protection + post-validation.
- “I do not know what prompt to write.” -> structured customization form + examples.
- “This output is too long/too formal/too generic.” -> length, audience, and intent controls.
- “I only dislike one phrase.” -> inline selection actions + alternatives.
- “The humanized version feels less academic.” -> context-aware humanizer with academic/professional variants and preservation checks.

3. Product Principles and Non-goals

3.1 Product principles

Principle	Meaning in UX / engineering
Simple first, advanced when asked	The default flow shows only text input + mode + generate. Customization appears behind “Sesuaikan”.
Document is the source of truth	After first generation, the user works in one editable document canvas; AI suggestions are previews until applied.
Selection over regeneration	If user selects a word/phrase/sentence, only that scope is sent to AI unless they explicitly choose a larger scope.
Original is never destroyed	Original text and applied AI versions are stored; restore creates a new version rather than deleting history.
Transparent transformation	Diff, local metrics, and concise AI change summaries explain what happened without exposing chain-of-thought.
Protected content is enforceable	Locks/citations are not merely prompt instructions; server-side validation checks the output.
AI only where AI adds value	No LLM for counts, diff, formatting, history, autosave, or standard UI interactions.
Indonesia-first writing quality	Indonesian academic/professional conventions are first-class behavior, not an afterthought.

3.2 Explicit non-goals for MVP

- Not a full office suite or Google Docs replacement.
- Not a plagiarism checker.
- Not an AI-detector bypass product and should not promise detector evasion.
- Not a translation product, even though Indonesian and English are supported for writing refinement.
- Not a team collaboration suite in MVP; real-time multi-user editing and comments can be later.
- Not a prompt playground. Users can add a short extra instruction, but the product should work without prompt expertise.

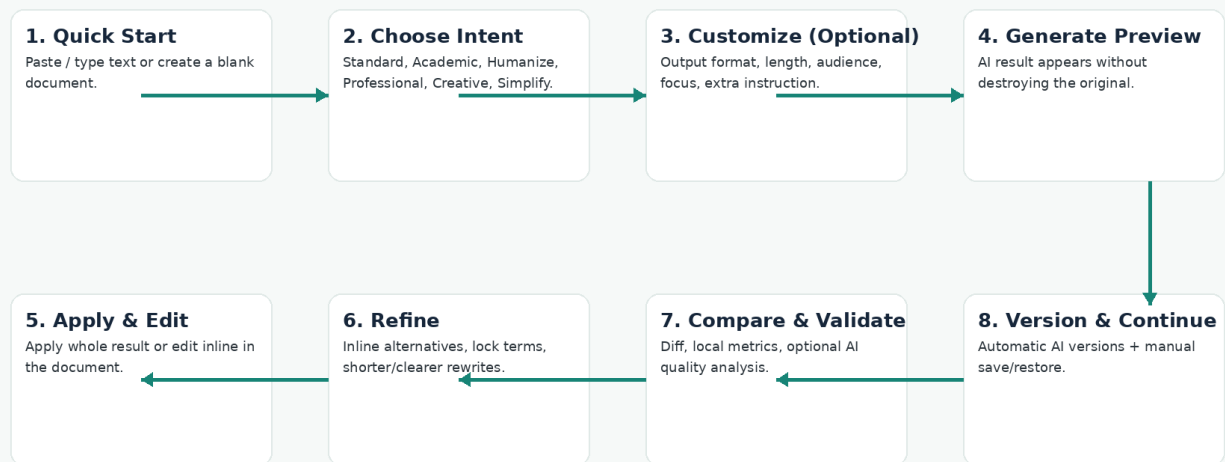
4. Complete Feature Inventory and Priority

Feature	Description	Priority	AI?
Writing Intent Engine	Mode selection that changes rewrite behavior: Standard, Academic, Humanize, Professional, Creative, Simplify.	MVP	Yes when generating
Structured Custom Writing Request	Output format, length, audience, focus, short extra instruction. Separate from mode selection.	MVP	Controls AI call; form itself no AI
Smart Rewrite Workspace	Google Docs-like editor with AI side panel and preview/apply behavior.	MVP	Mixed
Inline Rewrite / Alternatives	Select word/phrase/sentence and request alternative, clearer, shorter, etc.	MVP	Yes, scoped request
Term Lock / Protected Dictionary	Lock user-selected technical/domain terms.	MVP	No for lock; validation + AI instruction
Citation Protection	Recognize common citations and prevent unintended changes, especially in Academic mode.	MVP	Regex/local + validation
Diff / Compare	Inline or side-by-side comparison on demand.	MVP	No
Version Timeline	Original + AI-applied versions + manual checkpoints; restore safely.	MVP	No
Basic Writing Analytics	Words, characters, reading time, sentence count, change %.	MVP	No
AI Writing Quality Snapshot	Clarity, academic fit, naturalness/formality as heuristic scores.	MVP / optional call	Yes, on demand
Humanizer Change Insights	Concise explanation of AI-like patterns improved; preserve terms/citations.	MVP	Yes + local validation
Document upload DOCX/PDF	Import longer source files.	V1.1	Parsing mostly non-AI
Templates	Academic/professional starting templates.	V1.1	No
Export DOCX/PDF	Export final work.	V1.1	No
Collaboration / sharing	Multi-user comments/sharing.	Later	No/mostly backend
Plagiarism / AI detector	Separate product capability if validated later.	Later / separate decision	Potentially expensive

5. End-to-End User Journey

Blueprint 1 - End-to-End User Journey

Progressive disclosure: simple first, power tools when needed



5.1 First-time flow

7. User lands on the product and sees a large, familiar text entry card. The primary language is Bahasa Indonesia; language can be Auto, Bahasa Indonesia, or English.
8. User pastes/types text. The product immediately calculates local metrics (word count/character count) with no AI call.
9. User chooses a writing mode. Standard is the neutral default; Academic, Humanize, Professional, Creative, and Simplify are explicit alternatives.
10. User may click “Sesuaikan” to open the structured custom request form. This is optional and never blocks simple generation.
11. User clicks “Perbaiki Teks” / “Generate”. Only now is the semantic text sent to GPT-5.6 Luna.
12. AI result appears as a preview. The original text is still safe. The user can Apply, Compare, or Discard.
13. Applying the result creates a new AI version and opens/continues the document workspace.
14. From the workspace, user may edit manually, select a passage for scoped AI actions, lock terms, compare versions, or run optional writing quality analysis.

5.2 Returning-user flow

- Dashboard prioritizes recent documents and a quick-start composer.
- Resuming a document restores current working copy, active mode, locked terms, and version timeline.
- AI is never called merely because the document is opened.

5.3 Power-user refinement loop

- Select text -> inline action -> AI preview -> apply replacement -> automatic version event.
- Compare current with any previous version -> inspect diff -> restore or continue.
- Lock new terminology at any time -> all future AI transformations include and validate it.
- Open analytics -> local metrics appear immediately; AI quality snapshot runs only if requested.

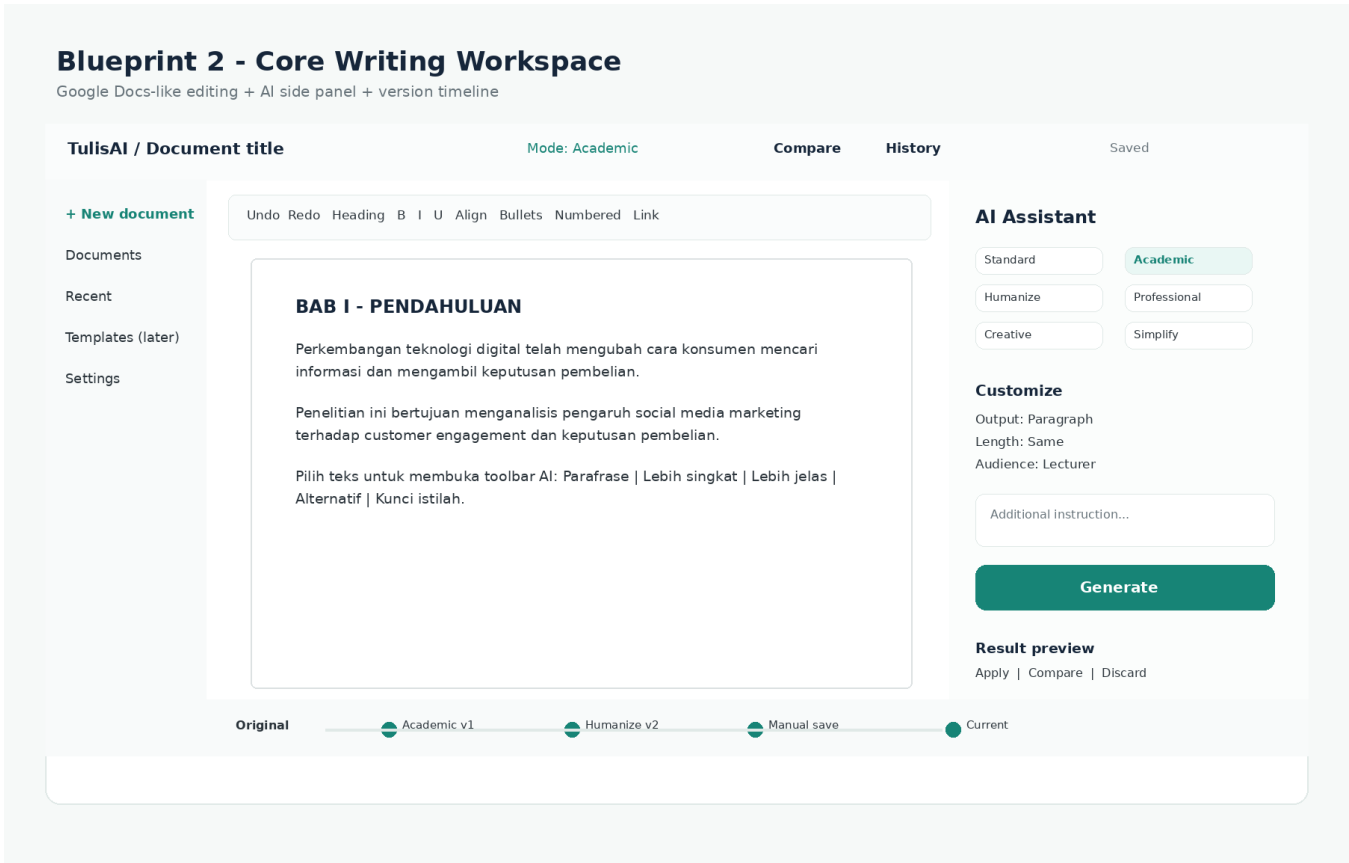
6. Information Architecture and Navigation

Route / surface	Purpose	Primary elements
/	Public landing / demo	Value proposition, quick demo, sign in/start CTA.
/app	Dashboard / quick start	Large input card, modes, optional customize, recent documents.
/documents/new	Focused new document flow	Text input/blank doc, language, mode, customize, generate.
/documents/:id	Core workspace	Editor, formatting toolbar, AI assistant, inline AI, bottom version timeline.
/documents/:id?panel=history	History drawer/state	Version timeline, rename, compare, restore.
/documents/:id?compare=a:b	Compare state	Diff between two versions; no editing until exit or restore.
/documents/:id?panel=analytics	Analytics drawer/state	Local metrics + optional AI quality analysis.
/settings	Settings	Language, default writing mode, privacy/preferences, account.

Navigation rule

Humanizer, Academic, Professional, Creative, and Simplify are modes/states of the same writing workspace, not disconnected mini-products. This keeps the user's document, versions, and locks in one continuous context.

7. UI Blueprint - Core Workspace



The mockup on the cover and this wireframe define the direction: a clean desktop-first document editor, persistent AI assistant on the right, lightweight navigation on the left, and a collapsible version timeline at the bottom. The center document remains the dominant visual element.

7.1 Workspace layout rules

- Main editor occupies the largest width; AI panel must not reduce the document to an unreadably narrow column.
- Right panel is collapsible. When closed, the editor expands.
- Permanent side-by-side original/output is avoided. Compare is an explicit mode so normal editing stays focused.
- Version timeline is bottom-fixed but collapsible; it communicates chronological drafts without technical Git terminology.
- Advanced controls are grouped and collapsed by default.

8. Page-by-Page UI/UX Specification

8.1 Public Landing Page

Objective

Explain the value quickly and let a new visitor understand that the product is an Indonesia-first AI writing workspace rather than a generic paraphraser.

Layout & hierarchy

- Top navigation: logo/working product name, Product, How it Works, Sign In, Start Writing.
- Hero: concise value proposition + large demo text box or screenshot.
- Mode preview chips: Academic, Humanize, Professional, Creative, Simplify.
- Visual section showing before/after, protected terms, version history, and diff.
- Trust section: “Your original is preserved”, “No prompt engineering required”, “Bahasa Indonesia first”.

Primary controls / buttons

- Start Writing
- Try Example
- Sign In

AI usage

- No LLM call on this page/state.

Deterministic / non-AI logic

- No user document is sent to AI until the user explicitly runs a demo or enters the app.
- Marketing carousel/animation can be static/client-side.

Important states

- Anonymous visitor
- Signed-in visitor
- Mobile layout collapses screenshots and keeps CTA visible.

Acceptance criteria

- Value proposition is understandable without scrolling.
- No misleading “AI detection bypass” claim.
- Primary CTA routes directly to quick start.

8.2 Authentication / First Onboarding

Objective

Get the user into the product with minimal friction; collect only preferences that materially improve the experience.

Layout & hierarchy

- Sign in / sign up form.
- After first sign-in, one lightweight preference screen: default language (Indonesia / English / Auto) and primary use case (Academic / Professional / General).
- Skip is allowed; defaults are safe.

Primary controls / buttons

- Continue
- Skip for now

AI usage

- No LLM call on this page/state.

Deterministic / non-AI logic

- Persist preferences in user profile.
- No AI call required.

Important states

- New account
- Returning account
- Authentication error
- Session expired

Acceptance criteria

- Onboarding can be completed in under one minute.
- User can change all preferences later.

8.3 Dashboard / Quick Start

Objective

Provide the familiar “big box in the middle” simplicity the user liked from QuillBot while retaining access to saved documents.

Layout & hierarchy

- Large centered composer: “Tulis atau tempel teks di sini”.
- Language selector: Auto / Bahasa Indonesia / English. Output language defaults to same as input.
- Writing mode chips directly under the text box.
- “Sesuaikan” link/button opens structured controls without leaving the page.
- Primary CTA appears only after sufficient text is present.
- Recent Documents below the fold with title, last edited time, active mode, and open action.

Primary controls / buttons

- Perbaiki Teks
- Sesuaikan
- Buat Dokumen Kosong
- Open Recent Document

AI usage

- Perbaiki Teks triggers the selected transformation.
- No AI call when selecting mode or filling customization.

Deterministic / non-AI logic

- Instant word/character count.
- Auto language detection can use a local heuristic/library; user can override.
- Recent document data from database.

Important states

- Empty input
- Text ready
- Generating
- Generation failed
- No recent documents

Acceptance criteria

- User can begin generation without understanding prompts.
- Advanced controls do not clutter the default state.

- Failure never clears the pasted text.

8.4 New Document / Quick Rewrite Surface

Objective

Create a clean first transformation and then transition into the richer workspace.

Layout & hierarchy

- Large editor/input area.
- Mode selector with Standard as default.
- Optional short sub-control based on mode, e.g. Academic context or Humanizer strength.
- Customize drawer/accordion.
- Generate button + scope summary ("Parafrase akademik, panjang sama, untuk dosen").

Primary controls / buttons

- Generate / Perbaiki
- Clear
- Paste
- Upload (V1.1)
- Customize

AI usage

- One transformation call using selected mode + structured custom request + protected terms (if any).

Deterministic / non-AI logic

- Input editing, paste, word count, language detection, button enable/disable.

Important states

- Valid input
- Too short
- Over plan limit
- Generating
- Preview ready
- AI error

Acceptance criteria

- Generation never overwrites input.
- The preview can be compared before applying.
- Applying creates the first AI version.

8.5 Writing Intent Selector

Mode	User promise	Default behavior	Typical sub-control
Standard / Paraphrase	Rewrite naturally while preserving meaning.	Balanced wording change; no forced register.	Change strength: Light / Balanced / Strong.
Academic	More academic, precise, and appropriate for thesis/research.	Citation protection on; terminology preservation emphasized.	Context: Thesis/Skripsi, Journal/Paper, General Academic.
Humanize	Make text feel natural and less formulaic while keeping the appropriate context.	Avoid robotic repetition; no deliberate errors; preserve meaning.	Context: Student/Academic, Professional, General; strength Light/Balanced/Strong.
Professional	Clear, concise, confident business writing.	Reduce filler; prioritize clarity and audience fit.	Audience or document type can be inherited from Customize.
Creative	More expressive, varied, engaging.	Allows larger stylistic variation but must not invent facts unless asked.	Creativity strength.
Simplify	Make complex text easier to understand.	Shorter sentences and simpler vocabulary.	Target audience/reading level through Customize.

Important interaction rule

Mode selection and Custom Writing Request are separate. The mode defines the transformation strategy; Custom Writing Request defines output shape, length, audience, and user-specific constraints.

8.6 Structured Custom Writing Request

Objective

Let users describe the desired result without writing a long system prompt or knowing prompt-engineering terminology.

Layout & hierarchy

- Opened by “Sesuaikan”. Use a side drawer or accordion, not a blank full-page prompt box.
- Output Format: Paragraph, Bullet Points, Numbered List, Table, Short Summary.
- Length: Shorter, Same, More Detailed.
- Audience: Lecturer/Academic Reviewer, Professional, Customer/Client, General Public, Other.
- Focus chips (optional multi-select): Clarity, Naturalness, Formality, Persuasiveness, Remove Repetition.
- Additional Request: short free-text field with example microcopy such as “Jangan ubah nama variabel” or “Fokus pada kesimpulan”.
- Summary line at bottom converts selections into plain-language request before generation.

Primary controls / buttons

- Apply Settings
- Reset
- Close

AI usage

- No AI while user fills the form. The final selections are compiled into the generation payload.
- Semantic transforms such as “turn this paragraph into meaningful bullet points” are executed by AI, not by a simple formatter.

Deterministic / non-AI logic

- Pure formatting changes (e.g., convert already-separated lines into bullet list) can be deterministic.
- Form validation and defaults are client-side.

Important states

- Default collapsed
- Customized
- Conflicting options (e.g., Shorter + More Detailed is prevented)
- Additional instruction too long

Acceptance criteria

- A user can get a tailored result without writing a prompt.
- Controls have sensible defaults and examples.
- Custom instructions are constrained in length to reduce cost and avoid prompt-like complexity.

8.7 Core Document Workspace

Objective

Provide the long-lived environment where the user edits, compares, refines, and versions the text.

Layout & hierarchy

- Left navigation: New Document, Documents, Recent; Templates later; Settings.
- Top bar: editable document title, current mode chip, language, Compare, History, autosave status.
- Formatting toolbar: Undo/Redo, paragraph style, bold/italic/underline, alignment, bullet/numbered list, link.
- Center: page-like rich text canvas.
- Right: AI Assistant panel, collapsible.
- Bottom: collapsible version timeline with Original and meaningful checkpoints.

Primary controls / buttons

- Compare
- History
- Save Version
- Open/Close AI
- New Document

AI usage

- Only explicit AI actions call GPT-5.6 Luna. Opening document, typing, formatting, saving, and switching versions do not.

Deterministic / non-AI logic

- Rich-text editing, autosave, undo/redo, local counts, diff rendering, version navigation.

Important states

- Saved
- Saving
- Offline/failed save
- AI panel open/closed
- Preview mode
- Compare mode

Acceptance criteria

- User never loses manually typed content when an AI call fails.
- Editor remains usable while AI panel is closed.
- Autosave does not create noisy version-history entries.

8.8 AI Assistant Panel

Objective

Centralize AI actions without turning the entire app into a chat interface.

Layout & hierarchy

- Mode tabs/chips at top.
- Scope indicator: Selection / Paragraph / Entire Document; selection is preferred when available.
- Mode-specific controls.
- Customize summary and quick edit.
- Generate/Rewrite button.
- Result preview card with Apply, Compare, Copy, Discard.
- Alternative suggestions appear in the same panel when requested.

Primary controls / buttons

- Rewrite / Generate
- Apply Result
- Compare
- Copy
- Discard
- Try Alternative

AI usage

- Every Generate/Try Alternative action is an explicit LLM call. Context is limited to selected scope plus minimal surrounding context and document metadata.

Deterministic / non-AI logic

- Copy, discard, apply-to-editor, compare, and panel navigation are non-AI.

Important states

- No selection
- Text selected
- Generating
- Preview ready
- Validation failed
- Provider error

Acceptance criteria

- Preview does not overwrite the document.
- Apply creates version metadata.
- The panel clearly states the scope being transformed.

8.9 Inline Text Interaction

Objective

Allow surgical improvements without full regeneration - one of the core differentiators.

Layout & hierarchy

- When text is selected, show a compact floating toolbar.
- Actions: Paraphrase, Shorter, Clearer, Alternatives, Humanize Selection, Lock Term, Custom.
- Clicking a changed phrase can show 3-5 alternative phrases in a small popover.
- Lock Term visually marks the protected span with a subtle indicator, not distracting highlighting.

Primary controls / buttons

- Replace
- Insert as Alternative
- Lock / Unlock
- Dismiss

AI usage

- Paraphrase/Shorter/Clearer/Alternatives/Humanize call AI with the selected span plus one sentence of context where needed.
- Batch 3-5 alternatives in one call rather than separate calls.

Deterministic / non-AI logic

- Selection detection, floating toolbar, lock/unlock, replacement, undo/redo are non-AI.

Important states

- Single word
- Phrase
- Sentence
- Multi-paragraph selection
- Locked selection

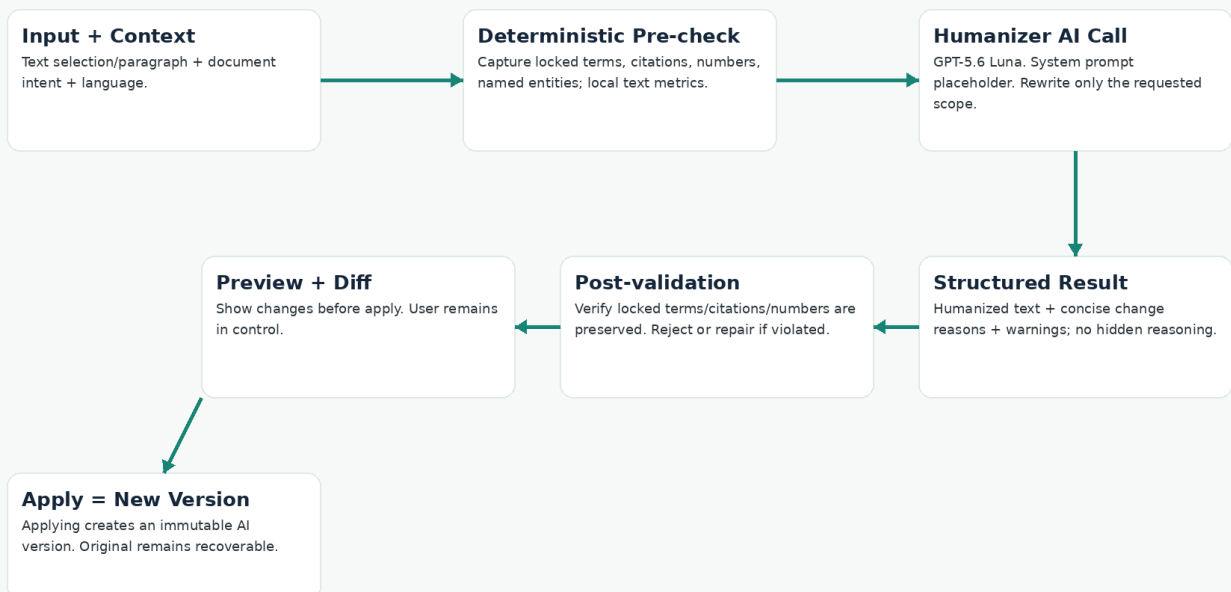
Acceptance criteria

- A small selection never causes the full document to be resent.
- Replacing text is reversible via undo and version history.
- Locked terms cannot be overwritten by an applied suggestion.

8.10 Humanizer Mode

Blueprint 3 - Humanizer Quality Pipeline

Natural writing, context-aware, with preservation guardrails



Objective

Produce natural, context-appropriate writing while preserving meaning, terminology, citations, and writing quality. Humanizer is a serious editing workflow, not an "AI detector bypass" gimmick.

Layout & hierarchy

- Humanize is selected as a mode in the same workspace.
- Context sub-choice: Student/Academic, Professional, General.
- Strength: Light, Balanced, Strong. Default Balanced.
- Advanced: Meaning Preservation preference (Conservative / Balanced / Flexible) may be exposed only when needed.
- Citation protection is automatically enabled for Academic context.
- Result preview includes "What changed" bullets and preservation warnings.

Primary controls / buttons

- Humanize
- Apply
- Compare
- Try More Natural
- Reduce Changes
- Discard

AI usage

- One primary humanization call should return humanized text + concise change categories in structured output.
- Optional deeper quality analysis is a separate on-demand call, not automatic.

Deterministic / non-AI logic

- Before the call: capture locked terms/citations/numbers. After the call: validate preservation; generate local diff/change %.
- No deliberately inserted grammar mistakes, typos, random punctuation, or fabricated personal anecdotes.

Important states

- Generating
- Preview valid
- Protected-term violation -> repair/reject
- No meaningful change needed
- User asks stronger/lighter humanization

Acceptance criteria

- Meaning is preserved according to selected strength.
- Recognized citations and locked terms remain unchanged.
- Humanizer does not reduce an academic text into casual language unless the user explicitly changes context.
- Output should improve naturalness through varied sentence rhythm, less generic phrasing, and context-appropriate word choice - not by adding mistakes.

8.11 Academic Safety: Term Lock + Citation Protection

Objective

Protect the exact spans users are most afraid AI will corrupt: theory names, research variables, technical terminology, brands, names, numbers, and citations.

Layout & hierarchy

- Lock by selecting text -> "Kunci Istilah".
- Protected Terms list in AI panel shows all active locks with remove action.
- Academic mode auto-detects common citation patterns and marks them as protected system spans.
- Optional suggested locks (acronyms/capitalized technical terms) are V1.1, not required for MVP.

Primary controls / buttons

- Lock Term
- Unlock
- View Protected Terms

AI usage

- LLM receives a compact list of protected terms and instruction not to modify them.

Deterministic / non-AI logic

- Regex/heuristics identify common citation patterns; post-generation validation compares required protected spans exactly.
- If violation occurs, result is not auto-applied. A bounded repair call may be attempted; otherwise show warning and retain original.

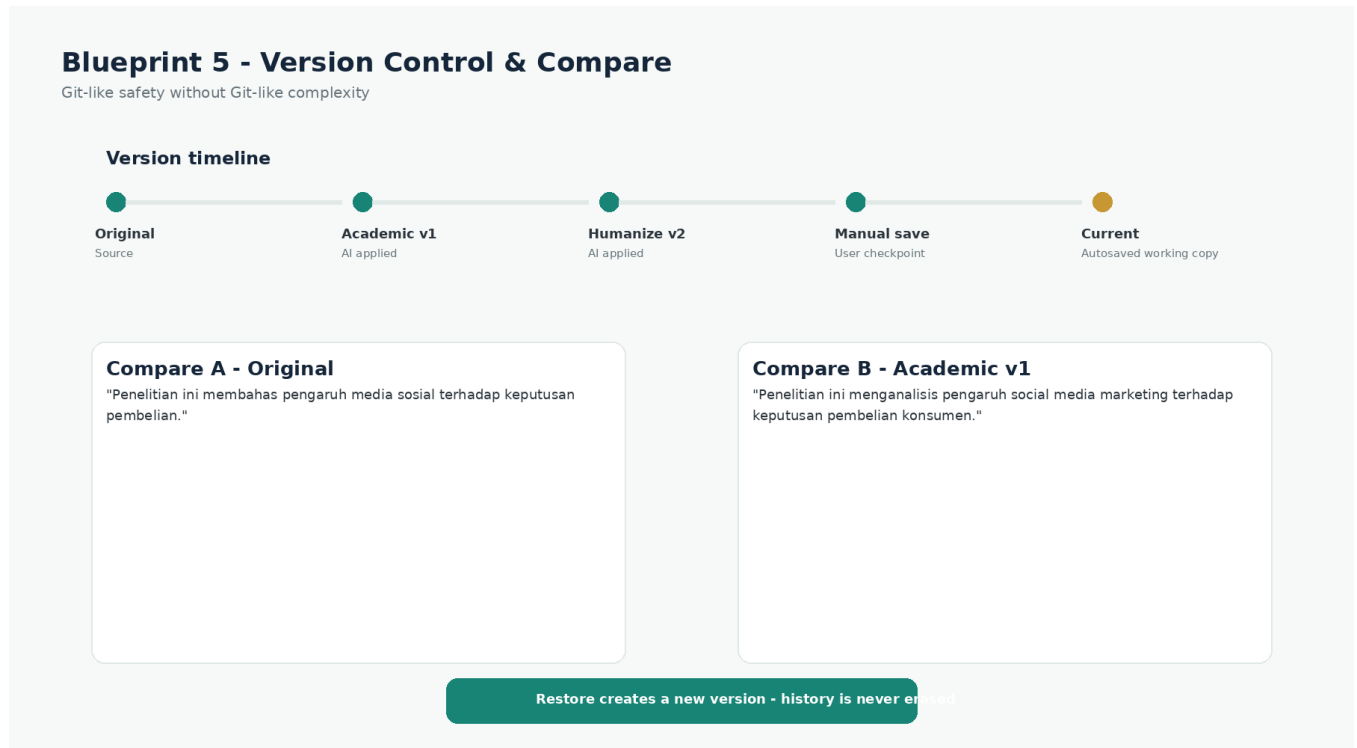
Important states

- No locks
- Manual locks
- Auto-protected citations
- Conflict where user requests translation/rewrite of a locked span

Acceptance criteria

- Manually locked strings are preserved exactly in applied AI results.
- Original remains unchanged if validation fails.
- User can unlock intentionally before requesting a change to the term.

8.12 Compare / Diff View



Objective

Give immediate confidence about what changed without forcing a permanent two-column editor.

Layout & hierarchy

- Compare is an explicit mode. Default view can be inline diff for readability; side-by-side is optional on wide screens.
- Version A and Version B selectors at top.
- Added text, removed text, and unchanged text have accessible visual treatments; do not rely only on color.
- Summary metrics: words added/removed, change %, reading time delta.

Primary controls / buttons

- Swap A/B
- Exit Compare
- Restore Version
- Use Version as Current

AI usage

- No LLM call on this page/state.

Deterministic / non-AI logic

- Text diff algorithm, change counts, percentages, render highlighting.

Important states

- No changes
- Small change
- Large change
- Compare Original vs Current
- Compare two AI versions

Acceptance criteria

- Compare uses no LLM.
- User can always identify which version is active.

- Restoring does not delete later history.

8.13 Version History / Timeline

Objective

Offer Git-like confidence in terminology normal users understand.

Layout & hierarchy

- Bottom timeline shows Original, applied AI versions, manual checkpoints, current working copy.
- History drawer provides timestamps, mode used, transformation scope, optional custom label.
- AI application automatically creates a version. Manual typing is autosaved but does not create a version every keystroke.
- Manual “Simpan Versi” creates a named checkpoint.

Primary controls / buttons

- Save Version
- Rename
- Compare
- Restore
- Duplicate (optional)

AI usage

- No LLM call on this page/state.

Deterministic / non-AI logic

- Immutable snapshot storage for meaningful versions; autosave storage for current working copy.

Important states

- Original immutable
- AI version
- Manual checkpoint
- Restored version
- Current autosave

Acceptance criteria

- Original can never be deleted from normal UI.
- Restore creates a new version based on the selected historical version.
- Timeline remains understandable without branch/commit vocabulary.

8.14 Writing Analytics / Validation Panel

Metric	How computed	AI cost	UX note
Word count	Local tokenizer	None	Always visible.
Character count	Local	None	Always visible / secondary.
Sentence count	Local sentence segmentation	None	Approximate for abbreviations.
Reading time	Words / configured reading speed	None	Marked estimate.
Change percentage	Diff algorithm	None	Shown after compare/rewrite.
Repeated words / phrases	Local frequency heuristics	None	Optional insight.
Clarity	AI qualitative evaluator	Low / on demand	Heuristic, not objective grade.
Academic fit	AI qualitative evaluator	Low / on demand	Only meaningful for Academic context.
Naturalness	AI qualitative evaluator	Low / on demand	Humanizer-oriented heuristic.
Formality	AI qualitative evaluator	Low / on demand	Display as descriptive band or score.

Objective

Provide fast evidence that text changed and optional qualitative feedback without spending tokens continuously.

Layout & hierarchy

- Persistent lightweight metrics in footer/status bar.
- Analytics drawer with Before/After cards.
- Local metrics populate instantly.
- Button “Analisis Kualitas Tulisan” runs AI qualitative assessment.
- AI scores are labeled “indikatif” and include 1-2 concise reasons, not a false objective grade.

Primary controls / buttons

- Analyze Writing Quality
- Compare Before/After
- Close

AI usage

- Only qualitative scores call AI and only when requested (or explicitly bundled by product plan later).

Deterministic / non-AI logic

- All counts, reading time, repeated words, and diff metrics local.

Important states

- No AI analysis yet
- Analysis loading
- Analysis ready
- Not applicable to selected mode

Acceptance criteria

- Opening analytics costs zero tokens.
- User understands which metrics are estimates/heuristics.

8.15 Upload / Import (V1.1)

Objective

Let academic users work with existing DOCX/PDF files without making import complexity block the MVP.

Layout & hierarchy

- Upload control on Dashboard/New Document.
- DOCX preferred for editable structure; PDF imported as text with clear limitations.
- Preview extracted text before creating document.

Primary controls / buttons

- Upload
- Cancel
- Create Document from File

AI usage

- No LLM call on this page/state.

Deterministic / non-AI logic

- File parsing, size/type validation, extraction, structure preservation where possible. AI is not required to read basic text.

Important states

- Unsupported file
- Scanned PDF
- Extraction warning
- Large file

Acceptance criteria

- Never silently alter source content during import.
- Scanned PDFs can be rejected or handled by a later OCR capability.

8.16 Settings

Objective

Persist sensible preferences without becoming a complex control center.

Layout & hierarchy

- Account profile.
- Default language: Auto/Indonesian/English.
- Default mode.
- Default humanizer context (optional).
- Privacy/data controls and document deletion.
- Usage/plan information.

Primary controls / buttons

- Save Settings
- Delete Document/Data where applicable

AI usage

- No LLM call on this page/state.

Deterministic / non-AI logic

- Profile persistence and validation only.

Important states

- Saved

- Save failed
- Account limits

Acceptance criteria

- Changing settings does not retroactively rewrite documents.
- Privacy language matches actual provider/storage behavior at launch.

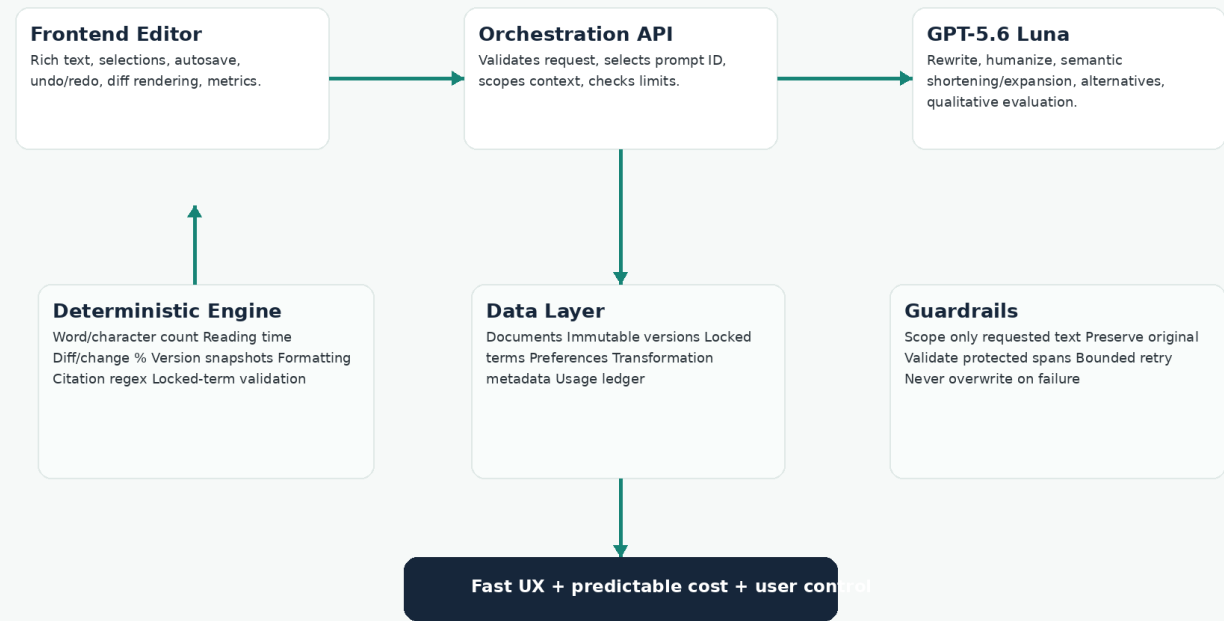
9. Cross-Page States: Empty, Loading, Error, and Recovery

State	Required UX behavior
Empty document	Show one primary action: start typing/paste. Secondary mode chips can remain visible.
AI generating	Do not freeze editor. Show generation status in AI panel; preserve current text. Cancel may be added if supported.
AI result ready	Clearly label as preview. Primary action Apply; secondary Compare, Copy, Discard.
Provider/API error	Keep input intact. Explain that generation failed and allow Retry. Never replace text with an error.
Validation failure	State which protected element changed ("1 locked term was modified"). Offer repair retry or discard.
Autosave failure	Show persistent "Not saved" indicator and local recovery buffer when feasible.
Too long for plan/context	Explain limit before calling AI; offer selected-text/paragraph rewrite or upgrade path later.
Language ambiguity	Default to Auto-detected language but show override before generation.
No meaningful rewrite	Tell user the text already fits the selected goal; allow stronger mode rather than forcing random changes.

10. AI vs Non-AI Decision Matrix

Blueprint 4 - Hybrid AI / Deterministic Architecture

Use AI only when semantic transformation is required



Action	AI?	Implementation decision
Word/character/sentence counts	No	Client/local deterministic functions.
Reading time	No	Local formula.
Diff / highlighted changes	No	Client or server diff algorithm.
Version history / restore	No	Database snapshots and application logic.
Manual formatting	No	Rich text editor commands.
Lock/unlock a term	No	Editor metadata + database.
Recognize common academic citations	Mostly no	Regex/rules for common formats; user can manually protect anything missed.
Validate locked terms/citations after AI	No	Exact/normalized comparison.
Paraphrase / rewrite	Yes	GPT-5.6 Luna.
Academic rewrite	Yes	GPT-5.6 Luna with Academic prompt ID.
Humanize	Yes	GPT-5.6 Luna with Humanizer prompt ID.
Professional/Creative/Simplify	Yes	GPT-5.6 Luna with corresponding prompt ID.
Meaningful paragraph -> bullet points	Yes	Semantic restructuring requires understanding.
Already-separated lines -> bullet formatting	No	Editor formatting command.
Shorter / expand while preserving meaning	Yes	Semantic transformation.
Inline phrase alternatives	Yes	Small scoped AI request.
Academic/naturalness/clarity qualitative score	Yes, on demand	Small structured evaluation request.
Copy/apply/discard result	No	UI/editor logic.

11. AI Orchestration Architecture

11.1 Model decision

MVP uses GPT-5.6 Luna as the primary generation model. The product architecture should still place the model behind an internal AI adapter/orchestration layer so prompts, validation, usage limits, and future routing can evolve without rewriting the editor.

11.2 Request context hierarchy

- 15. Always send the smallest user-requested scope: selected phrase -> selected sentence -> paragraph -> full document only when explicitly requested.
- 16. Include minimal neighboring context needed to preserve coherence (for example one sentence before/after).
- 17. Include compact metadata: language, selected writing mode, customization fields, protected terms/citations relevant to the scope.
- 18. Do not resend the entire version history. Use the currently active content and compact preferences only.

11.3 Structured output requirement

Field	Purpose
transformed_text	Text to preview/apply.
change_categories	Short labels such as “reduced repetition”, “more academic wording”, “shortened sentences”.
warnings	Potential ambiguity or inability to satisfy conflicting request.
protected_terms_seen	Optional acknowledgement used for debugging/telemetry; server validation remains authoritative.
alternatives[]	Only for inline alternative request.
quality_scores{}	Only for explicit evaluator call.

11.4 AI application guardrails

- Never directly write LLM output into the document before showing a preview or explicit apply action, except if the user later enables a deliberate “instant replace” preference.
- Bound retries. For MVP, at most one repair retry after a protected-term/citation validation failure; otherwise surface the failure and keep original text.
- No chain-of-thought is requested or stored. Only concise result metadata/change categories are needed.
- Prompt injection inside user text is treated as content, not as system instruction; server-defined system behavior has precedence.
- If a user request conflicts with a locked term, ask them to unlock it rather than silently ignoring the lock.

12. System Prompt Registry - Placeholders Only

Per product decision, the system-prompt content is intentionally not authored here. This section defines what prompts must exist so they can be researched independently later without changing the product architecture.

Prompt ID	Purpose	System prompt content	Required inputs	Expected output
P01_STANDARD_REWRITE	Neutral paraphrase	[TBD - user research]	text, language, strength, custom_request, protected_terms	transformed_text + change_categories + warnings
P02_ACADEMIC	Academic rewrite	[TBD - user research]	text, academic_context, language, custom_request, protected_terms, citations	transformed_text + change_categories + warnings
P03_HUMANIZER	Context-aware humanization	[TBD - user research]	text, humanizer_context, strength, preservation, protected_terms, citations	humanized_text + change_categories + warnings
P04_PROFESSIONAL	Professional rewrite	[TBD - user research]	text, audience, length, custom_request	transformed_text + categories
P05_CREATIVE	Creative rewrite	[TBD - user research]	text, creativity_strength, audience, custom_request	transformed_text + categories
P06_SIMPLIFY	Simplification	[TBD - user research]	text, audience, length, format	transformed_text + categories
P07_INLINE_ALTERNATIVES	Phrase/sentence alternatives	[TBD - user research]	selected_text, local_context, mode, protected_terms	3-5 alternatives
P08_CUSTOM_TRANSFORM	Structured custom request	[TBD - user research]	text + normalized form selections	transformed_text + categories + warnings
P09_QUALITY_EVAL	Qualitative writing analysis	[TBD - user research]	text, mode, language	compact score bands + short reasons
P10_REPAIR	Repair protected-span violation	[TBD - user research]	failed_output, required spans, original scope	corrected_text only

13. Humanizer - Detailed Product Specification

Humanizer is one of the product's hero workflows and therefore needs stronger quality guarantees than a generic “make it human” prompt. Its objective is natural, context-appropriate language, not artificial randomness.

13.1 Humanizer quality definition

- Preserve the original factual meaning unless the user explicitly asks to expand/alter content.
- Reduce repetitive sentence openings and repetitive syntactic patterns.
- Replace generic or overly formulaic transitions with context-appropriate phrasing when helpful.
- Vary sentence rhythm naturally without making the text erratic.
- Use vocabulary appropriate for the selected context; academic humanization remains academically appropriate.
- Do not intentionally add typos, grammar errors, strange punctuation, fabricated anecdotes, or meaningless filler.
- Do not advertise or optimize around defeating AI detectors. The product optimizes writing quality and naturalness.

13.2 Humanizer controls

Control	Default	Behavior
Context	Inherited from document; General if unknown	Student/Academic, Professional, General.
Strength	Balanced	Light changes fewer patterns; Strong permits larger restructuring but still preserves meaning.
Meaning preservation	Balanced / hidden in advanced	Conservative restricts structural change; Flexible allows larger rewrite. Avoid fake percentage precision.
Output format	Same as current	Can be overridden through Custom Writing Request.
Protected terms	Always enforced	User locks + detected citations in academic context.

13.3 Humanizer MVP pipeline

19. Pre-check locally: capture protected terms, citations, numbers, basic paragraph/sentence metrics.
20. Build a compact AI request from selected scope + context controls + protected spans.
21. One primary GPT-5.6 Luna call returns humanized text and concise change categories.
22. Validate protected spans deterministically. If invalid, run one bounded repair or reject preview.
23. Render local diff and change percentage.
24. User compares and explicitly applies. Apply creates a new version.
25. Optional qualitative “Naturalness” analysis is not required for every humanizer call; it is on demand to control cost.

13.4 Humanizer acceptance criteria

- 100% of manually locked terms are present exactly in an applied result unless user unlocked them.
- Recognized citations are preserved in Academic context.
- No new factual claims are introduced in normal Humanizer mode.
- User can see a diff before applying.
- If strong humanization causes unacceptable changes, user can request “Reduce Changes” without restarting the document.
- Humanizer operates on a selected passage when selected; it must not resend the full document by default.

14. Academic Mode - Detailed Product Specification

14.1 Objective

Academic mode should improve clarity, precision, and academic register for Indonesian/English academic writing while respecting research terminology and citations. It must not simply replace common words with unnecessarily complicated synonyms.

14.2 Default academic protections

- Citation protection on.
- Locked terms always enforced.
- No invented references, authors, years, data, or claims.
- No automatic change of variable names, theory names, abbreviations, brand names, or defined constructs when locked.
- Meaning-preservation preference defaults conservative/balanced.

14.3 Academic subcontexts

Subcontext	Style target
Skripsi / Thesis	Clear, formal, academically appropriate, not unnecessarily ornate.
Journal / Paper	More concise and publication-oriented while preserving evidence and citations.
General Academic	Neutral academic register for assignments/reports.

14.4 Academic quality snapshot

The optional AI evaluator can report Clarity, Academic Fit, and Formality as heuristic bands/scores. The UI must state that these are writing-assistance indicators, not academic grades or plagiarism measures.

15. Data Model and Versioning

Entity	Key fields	Notes
User	id, language_preference, primary_use_case, plan/usage metadata	Avoid storing unnecessary personal content.
Document	id, owner_id, title, source_language, current_content_rich, current_mode, current_preferences, created_at, updated_at	Current working copy is autosaved.
DocumentVersion	id, document_id, parent_version_id(optional), snapshot_rich, label, source_type(original/AI/manual/restore), mode, created_at	Immutable meaningful checkpoint.
AITransformation	id, document_id, version_before, version_after(optional), prompt_id, scope_type, scope_hash, mode, normalized_controls, status, token/usage metadata	Do not store hidden reasoning.
LockedTerm	id, document_id, exact_text, normalized_text(optional), created_at, source(manual/system)	Manual locks authoritative.
ProtectedCitation	document_id, span/text, detection_type	Can be derived dynamically rather than always persisted.
UsageLedger	user_id, transformation_id, model, input/output units, timestamp	Used for cost and plan limits.

15.1 Autosave vs version snapshot

- Autosave keeps the current document safe but should not create timeline noise.
- AI Apply always creates a meaningful version snapshot.
- Manual “Save Version” creates a meaningful snapshot.
- Restore creates a new snapshot derived from an older version; history remains immutable.
- Original source version remains identifiable and protected from normal deletion.

16. High-Level API / Service Contracts

Endpoint / service	Purpose	LLM?	Key behavior
POST /transform	Full/paragraph/selection rewrite	Yes	Accept normalized mode/custom controls, protected spans, scope; return preview payload.
POST /transform/alternatives	Inline alternatives	Yes	Small context only; return 3-5 suggestions.
POST /analyze-quality	Qualitative writing scores	Yes	On-demand; compact structured output.
POST /validate-protected	Server-side protection check	No	Compare output vs required protected spans.
GET/PUT /documents/:id	Load/autosave current document	No	Optimistic concurrency/recovery.
POST /documents/:id/versions	Create manual/AI snapshot	No	Immutable snapshot.
GET /documents/:id/versions	History list	No	Paginated if needed.
POST /documents/:id/restore	Restore prior version as new current snapshot	No	Never destroys newer history.

16.1 Transform request example - conceptual

Payload shape (not implementation syntax)

scope: selection|paragraph|document; text; language; mode; mode_settings; custom_request {format,length,audience,focus,extra}; protected_terms[]; protected_citations[]; context_before/context_after (minimal).

16.2 Transform response example - conceptual

Response shape

transformed_text; change_categories[]; warnings[]; validation_status; usage_metadata. Alternatives and quality scores are separate request types unless explicitly requested.

17. Cost and Latency Optimization Plan

Technique	Product behavior	Impact
Smallest-scope AI calls	Selection/paragraph by default; full document only explicitly.	Major token reduction.
Progressive customization	No LLM while user changes dropdowns/chips.	Zero wasted calls.
No AI per keystroke	AI actions are explicit buttons.	Predictable cost and less latency.
Batch alternatives	Request 3-5 options in one inline call.	Avoid repeated round trips.
Deterministic analytics first	Counts/diff/history/reading time local.	Zero token cost for frequent feedback.
On-demand qualitative scoring	AI score only when user opens/runs analysis.	Avoid automatic evaluation after every edit.
Compact metadata	Send mode IDs/normalized controls and only relevant protected spans/context.	Lower prompt overhead.
Bounded retries	Only one repair retry on validation failure.	Prevents runaway cost.
Reuse generated previews	Do not regenerate when user simply reopens compare/history.	No duplicate calls.
Usage ledger	Measure cost per successful applied rewrite, not calls alone.	Enables pricing decisions.

17.1 Cost KPI

- Cost per successful applied transformation.
- Average AI calls per active document session.
- Percentage of generations applied vs discarded.
- Percentage of refinements handled by selection-level calls vs full-document calls.
- Repair retry rate caused by protected-span violations.

18. UI Microcopy and Interaction Language

Context	Recommended Indonesian copy
Main input placeholder	Tulis atau tempel teks yang ingin kamu perbaiki...
Mode heading	Kamu ingin teks ini menjadi seperti apa?
Customize CTA	Sesuaikan hasil
Generate CTA	Perbaiki Teks
Humanizer CTA	Buat Lebih Natural
Preview primary	Gunakan Hasil Ini
Preview secondary	Bandingkan
Discard	Buang Hasil
Term lock	Kunci Istilah
Protected list	Istilah yang tidak boleh diubah
History	Riwayat Versi
Manual checkpoint	Simpan Versi
Analytics	Analisis Tulisan
AI analysis CTA	Analisis Kualitas Tulisan
Validation warning	Hasil ini mengubah istilah yang dikunci. Teks belum diterapkan.

Tone of UI copy
Use plain Indonesian, not AI jargon. Avoid “prompt”, “system message”, “temperature”, “token”, “branch”, or “commit” in user-facing UI.

19. Product Analytics and Success Metrics

19.1 Funnel events

Event	Why it matters
quick_start_text_entered	Measures entry activation.
mode_selected	Mode demand distribution.
customize_opened	Whether structured controls are valuable without being mandatory.
generation_started / completed / failed	Reliability and conversion.
result_applied / discarded	Core quality signal.
compare_opened	Trust/validation behavior.
inline_action_used	Differentiator adoption.
term_locked	Academic safety adoption.
version_restored	Shows value of reversible history and potential quality issues.
humanizer_generated / applied	Hero workflow quality.
quality_analysis_requested	Demand for AI evaluation.

19.2 Product KPIs

- Activation: user applies a first useful transformation.
- Apply rate: percentage of completed AI previews applied.
- Regeneration rate: excessive full regenerations should decrease as inline/custom controls improve.
- Inline refinement adoption.
- Return rate to saved documents.
- Humanizer apply rate and “reduce changes” rate.
- Protected-span violation rate (target should approach zero for applied outputs).
- Cost per applied result and latency distribution.

19.3 Privacy rule for analytics

Product analytics should log interaction metadata, not raw document text. Full text should not be copied into general analytics/event systems. AI transformation logging should be separated and governed by product privacy requirements.

20. Quality Assurance and Acceptance Test Plan

Area	Critical tests
Standard rewrite	Meaning preserved; selected mode followed; original preserved; preview before apply.
Academic	Citations preserved; locked terms preserved; no fake references/data; appropriate Indonesian/English register.
Humanizer	Naturalness improves without grammar degradation; context maintained; protected spans preserved; no deliberate errors.
Custom format	Paragraph -> meaningful bullets/table/summary follows requested structure; conflicting controls prevented.
Inline alternatives	Only selected scope changes; surrounding context remains untouched.
Diff	Added/removed spans correct across punctuation and Unicode; no AI call.
Versioning	Original immutable; AI apply creates snapshot; restore creates new snapshot; autosave does not flood history.
Failure recovery	Provider failure, validation failure, and save failure never erase user content.
Language	Indonesian and English transformations respect selected language; Auto can be overridden.
Cost	No AI on opening docs, typing, formatting, compare, history, local analytics.

20.1 Golden evaluation set before launch

- Build a curated Indonesian dataset covering thesis introduction, literature review, methodology explanation, discussion, business email, report, marketing copy, and mixed technical terminology.
- For each sample, define protected terms/citations and expected constraints rather than asking reviewers for subjective “better” only.
- Human review rubric: meaning preservation, language naturalness, mode fit, grammatical quality, terminology preservation, unnecessary verbosity, unsupported claims.
- Use the same evaluation set when system prompts change so improvements are measurable.

21. Security, Privacy, and Data Handling Requirements

- Documents are private by default.
- Use transport encryption and encrypted persistent storage appropriate to the chosen stack.
- Provide document deletion and account data controls.
- Do not place raw user documents in generic analytics logs.
- Define and document the AI provider's actual retention/training configuration before launch; product copy must match reality.
- System prompts and internal model instructions stay server-side.
- Input text may contain prompt-like instructions; treat them as user content unless they come from the structured product controls.
- Rate limits, file-size limits, and abuse controls should protect both cost and availability.

22. Responsive and Accessibility Requirements

- Desktop is the primary full-workspace target because long-form writing and compare/version controls need space.
- Tablet should support editing with the AI panel as an overlay/drawer.
- Mobile MVP can prioritize quick rewrite, preview, and basic editing; complex compare/version management may be simplified.
- Keyboard navigation for editor toolbar, AI actions, and version timeline.
- Diff must not rely on green/red color alone; use underline/strikethrough or labels/icons.
- Focus states visible; buttons have text labels or accessible names.
- Do not use tiny fixed text in side panels; body UI target should remain comfortably readable.

23. MVP Scope and Release Plan

Phase	Must deliver
MVP Foundation	Auth, Dashboard Quick Start, document entity/autosave, Indonesian/English support, Standard/Academic/Humanize/Professional/Creative/Simplify modes, structured customization, GPT-5.6 Luna orchestration.
MVP Differentiators	Inline rewrite/alternatives, term lock, citation protection for common academic patterns, result preview/apply, diff compare, version timeline, local metrics.
MVP Humanizer Quality	Context-aware humanizer, strength control, protected-span validation, change summary, bounded repair, no "detector bypass" positioning.
MVP Optional / if capacity	On-demand AI writing-quality snapshot.
V1.1	DOCX/PDF import, export, templates, suggested term locks, more polished mobile behavior.
Later	Collaboration/sharing, team/university workspaces, additional models/routing, separate plagiarism/detector decision.

23.1 Suggested implementation sequence

26. Build rich-text document model, autosave, and immutable version snapshots first.
27. Build deterministic local tools: counts, diff, locks, citation recognition, compare UI.
28. Add AI orchestration with one Standard prompt placeholder and preview/apply flow.
29. Add the remaining mode prompt IDs without changing UX architecture.
30. Add structured Customize and normalize it into the same transform API.
31. Add selection-level actions and alternatives.
32. Add Academic protection validation and repair flow.
33. Add Humanizer-specific context/strength and quality guardrails.
34. Add on-demand qualitative analytics last; it is useful but not required to prove the core editing loop.

24. Risks and Mitigations

Risk	Why it matters	Mitigation
Humanizer becomes “worse writing”	Some products fake human style with errors.	Quality rubric forbids deliberate errors; context-specific humanization; preview/diff.
Too many controls overwhelm new users	Power features can hurt activation.	Progressive disclosure; default mode + one Generate button; Customize collapsed.
Academic terminology changes	High-trust failure.	Manual term locks + citation detection + deterministic output validation.
Full-document calls become expensive	Long thesis documents can be costly.	Selection/paragraph scope first; explicit full-document action; plan limits.
Version history becomes noisy	Autosave can create hundreds of versions.	Separate autosave from meaningful snapshots.
AI scores appear authoritative	Users may over-trust heuristics.	Label as indicative and explain dimensions; no plagiarism/detector claims.
Bilingual behavior drifts	Indonesian quality may lag English.	Indonesia-first evaluation set and prompt research; language-specific QA.
UI resembles existing competitors too closely	Weak differentiation.	Entry is familiar, but core workspace, inline refinement, locks, diff, and versions create distinct workflow.

25. Open Decisions - Intentionally Not Locked Yet

Decision	Why still open	When to decide
Final product name/brand	No name selected in discussion.	Before visual identity/landing implementation.
Exact system prompt text for every mode	User explicitly wants to research prompts separately.	Before AI QA / implementation freeze.
Pricing/plan limits	Requires cost measurements from real token usage.	After prototype telemetry.
Exact rich text editor library	Technical choice, not product requirement.	Engineering design phase.
Exact export/import fidelity	Depends on editor/data model.	V1.1 planning.
Team collaboration	Not needed for MVP.	After single-user retention is proven.

What is NOT open

The core workflow, mode structure, separate structured customization, Google Docs-like workspace, inline refinement, term locks, diff, version history, hybrid AI/non-AI approach, Humanizer quality principles, and GPT-5.6 Luna as the MVP model are already defined in this PRD.

Appendix A. Feature-to-User-Value / Cost Matrix

Feature	User value	LLM/server cost	Frontend complexity	Backend complexity
Mode-based rewrite	High	Medium per generation	Low	Medium
Structured Customize	High	Low overhead	Medium	Low/Medium
Inline alternatives	High	Low per call	Medium	Low/Medium
Term lock	Very High academic	Very Low	Medium	Low
Citation protection	High academic	Very Low	Low/Medium	Medium validation
Diff viewer	High	None AI	Medium	Low
Version history	High	No LLM; storage only	Medium	Medium
Local analytics	Medium/High	None AI	Low	Low
AI quality snapshot	Medium/High	Low, on demand	Low	Medium
Humanizer	Very High	Medium	Medium	Medium/High due validation
DOCX/PDF import	High for academics	No LLM for normal parse	Medium	Medium

Appendix B. UX Checklist for Design Handoff

- Design Quick Start desktop screen with large central input and mode chips.
- Design collapsed/expanded Structured Customize drawer.
- Design Workspace desktop layout using the blueprint in this PRD.
- Design selected-text floating AI toolbar.
- Design result preview card with Apply / Compare / Copy / Discard.
- Design protected-term visual state and Protected Terms list.
- Design Humanizer controls and preservation warning state.
- Design bottom version timeline and expanded History drawer.
- Design Compare mode with inline diff and optional side-by-side wide-screen layout.
- Design Analytics drawer with local metrics and optional AI analysis CTA.
- Design loading, error, validation failure, offline save, and “no meaningful change” states.
- Keep Indonesian microcopy simple and avoid prompt/LLM/token/Git terminology.

Appendix C. Engineering Definition of Done for MVP

- Every semantic AI action maps to a documented prompt ID and structured request schema.
- Every AI preview is reversible and never silently destroys the original.
- No AI request is triggered by typing, formatting, opening history, diff, or local analytics.
- Locked terms are deterministically validated before an AI result can be applied.
- Academic citation protection is covered by regression tests for supported patterns.
- Humanizer regression set covers Indonesian academic, professional, and general text.
- Applied AI transformations create version snapshots with mode/scope metadata.
- All core UI states have explicit loading/error/recovery behavior.
- Product analytics measures apply/discard and cost per applied transformation without sending raw document text to general analytics.
- Prompt content can be replaced without changing the editor or transform API contract.

END OF PRD

This document is the consolidated blueprint for product, UI/UX, AI, and engineering implementation.

System prompt contents remain intentionally TBD for separate research.